



APPLICATION FOR ETHICS APPROVAL

Mae'r ddogfen hon hefyd ar gael yn Gymraeg / This document is also available in Welsh

When undertaking a research or innovation project, Cardiff Met staff and students are obliged to complete this form in order that the ethics implications of that project may be considered.

The document **Ethics application guidance notes** will help you complete this form and is available from the Ethics Governance Section of the Cardiff Met website. The School or Unit in which you are based may also have produced some guidance documents which you can access via your supervisor or School Ethics Coordinator.

PLEASE NOTE:

Participant recruitment or data collection MUST NOT commence until ethics approval has been obtained.

Completion of all sections of Part One and the Declaration is compulsory in all cases.

Forms which have been modified to delete or edit any of the questions will be returned to the applicant without being considered.

PART ONE

1A GENERAL INFORMATION	
Name of applicant:	Dissanayake Thalangama Appuhamilage Dilaksha Haritha Dissanayake
Supervisor (if student project):	Ms. Upeka Wijesinghe
School / Unit:	ICBT Campus / Cardiff Metropolitan University
Student number (if applicable):	GM/BSCSD/06/22 (Cardiff: 20360315)
Programme enrolled on (if applicable):	BSc Software Engineering
Project Title: If using a working title, it should convey what the project is about	Resilient Multimodal Intelligence: An AI-Driven Lost and Found Matchmaking System with Context-Aware Graceful Degradation
Expected start date of data collection:	19 June 2026
Approximate duration of data collection:	2 weeks
Funding Body (if applicable):	N/A (Self-funded)
Other researcher(s) working on the project: If your collaborators are external to Cardiff Met, include details of the organisation they represent	

1B SECURITY SENSITIVITY	
Will the project have security sensitive implications? NB: Any project which falls under this definition will be referred to the Prevent Co-ordinator by the Committee which is considering the application so it can be risk assessed in line with the Prevent Policy . Ethics approval will only be granted once the Prevent Co-ordinator is content	No
If yes, please explain what they are and the measures that are proposed to address them. (500 words max)	



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1C NON-TECHNICAL SUMMARY

Please provide a non-technical summary of the project below (max 250 words)

This project aims to develop an intelligent mobile application that helps reconnect people who have lost personal belongings with individuals who have found them. Current lost and found processes often rely on social media posts, messaging groups, and manual searching, which can be inefficient and fragmented.

The proposed system will use artificial intelligence, computer vision, and location-based analysis to improve the matching process between lost and found items. Users will be able to submit descriptions and photographs of items, and the system will automatically identify potential matches.

The project also focuses on user privacy and security. Personal contact information will not be publicly displayed, and communication between users will be facilitated through secure in app messaging. The research will involve collecting feedback through a questionnaire and conducting user acceptance testing to evaluate usability, usefulness, and user satisfaction.

The findings will be used to assess the effectiveness of the proposed system and identify areas for future improvement.

1D DATA COLLECTION AND STORAGE

What types of data will be collected or created as part of the project? (max 250 words)

The project will collect questionnaire responses, user acceptance testing (UAT) feedback, user account information, uploaded images of lost and found items, descriptive text data, and approximate location information used for matching purposes. Survey responses will be anonymised and used only for academic research purposes. No sensitive personal information will be intentionally collected beyond what is necessary for system functionality and research evaluation.

How will you manage access to and security of the data? (max 500 words)

Data will be housed in a secure PostgreSQL database with spatial extensions. All API communications between the mobile application, the Laravel API Gateway, and the Python microservice will be encrypted and handled via secure token based authentication (Laravel Sanctum). Location data will be masked, and real-time chat sessions will preserve user anonymity.

Will the project involve human samples and/or human cell lines?

No

Will the data collected be subject to the data retention protocols of either:

- Human Tissue Authority (HTA)
- The Health Research Authority - Health and Care Research Wales (HRA-HCRW)

Yes ☐

For any project which is subject to the data retention protocols of either the HTA or HRA-HCRW, you are required to develop a data storage plan to be submitted alongside your ethics application. The plan should expand on the answers you have provided to the preceding two questions and should show how it will meet the expectations of the external body involved.

No ☒

Please confirm that the data collected will be stored in a manner which complies with Cardiff Met requirements via one of the following statements.

STATEMENT 1: FOR STUDENTS ON TAUGHT COURSES



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I confirm that any non-anonymised data related to research participants will only be stored on OneDrive, or by agreement with supervising staff, on Figshare, and that all data held elsewhere will be deleted, unless it is anonymised.	☒
STATEMENT 2: FOR STAFF APPLYING ON BEHALF OF STUDENTS ON TAUGHT COURSES I confirm that all students covered by this application are aware of their obligation to ensure that non-anonymised data related to research participants must only be stored on their Cardiff Met student OneDrive account and that all data held elsewhere must be deleted, unless it is anonymised.	☐
STATEMENT 3: FOR RESEARCH STUDENTS AND STAFF I confirm that any non-anonymised data related to research participants will be stored in a secure manner (using a platform such as OneDrive or FigShare) and that all data held elsewhere will be deleted unless it is anonymised.	☐

1E PROJECT CATEGORIES	
From the options provided, select the category which best describes your project, bearing in mind that “desk based” can encompass any non-practice based project which takes place primarily outside of a laboratory.	Desk based ☒ Laboratory based ☐ Practice based ☐
Do any of the following statements describe your project?: - Desk based involving only documents and NOT involving the collection of data from participants - OR - Laboratory based NOT involving human participants, human samples, animals or animal derived material - OR - Practice based (eg curatorial, practice audit) NOT involving human participants	No
Answering YES indicates that the project does not include any participants and you will not therefore be collecting participant data. If this is the case, complete the Declaration at the bottom of the form and forward this form to your School Ethics Committee (or equivalent). No further information regarding your project is required and you do not need to complete any more sections of this form. If you have answered NO , please proceed to the next question.	

1F NHS	
Does your project involve the NHS? If no, proceed to 1G	No
If yes, has the NHS deemed that your project requires an IRAS application?	
Yes ☐ For a project which requires an IRAS application, you must submit the following alongside your ethics application. - The study protocol document - The completed IRAS form - Any other documents relevant to the IRAS application	



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Failure to do this will result in your application being returned to you.

No ☒

Proceed to Part Two (below)

1G ETHICS APPROVAL OBTAINED ELSEWHERE

Has your project already received ethics approval from a body or organisation other than Cardiff Met?

Yes ☐

No further information regarding your project is required.

Complete the Declaration at the bottom of this form and submit it to your School Ethics Committee (or equivalent) alongside the following documents:

- Copy of the externally approved ethics application and any associated documents.
- Copy of the approval letter from the external organisation.

No ☒

Proceed to Part Two (below)



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PART TWO

RESEARCH DESIGN	
1 Will you be using an approved protocol in your project? (An approved protocol is one which has been approved by Cardiff Met to be used under supervision of designated members of staff. For details of protocols in use in your School or Unit, contact your Ethics Coordinator)	No
If yes, please state the name and code of the approved protocol to be used	
2 Describe the research design to be used in your project by giving details under the following four headings. If your project involves the use of an approved protocol, much less detail will be required but you should indicate which areas of the project are covered by the protocol • The research method(s) to be used; • Sample and sampling; • The proposed participant group, including recruitment methods, activities to be undertaken, the expected time commitment and details of any proposed payments; • Analytical techniques to be used. (max 1000 words)	
This project will adopt a mixed-methods approach consisting of a questionnaire survey, system development, and User Acceptance Testing (UAT). The questionnaire will be conducted to gather requirements and understand user experiences with existing lost and found processes. Following system development, UAT will be conducted to evaluate usability, functionality, and user satisfaction. Sample and Sampling Participants will be recruited using convenience sampling. Approximately 50-150 participants will be invited to complete the questionnaire, while 10-20 participants will be invited to participate in UAT activities. Participant Group, Recruitment and Activities Participants will consist of university students, staff members, and members of the public aged 18 years or above. Recruitment will be conducted through university networks, social media platforms, and personal contacts. Participation will be entirely voluntary, and no financial incentives will be offered. The questionnaire is expected to take approximately 5 -10 minutes to complete. UAT activities are expected to require approximately 30-40 minutes. Analytical Techniques Survey responses will be analysed using descriptive statistical techniques such as percentages, frequencies, and charts. UAT feedback and qualitative comments will be analysed using thematic analysis to identify common usability issues and improvement opportunities. System performance metrics such as matching accuracy and response times may also be evaluated.	
3 Will the project involve deceptive or covert research?	No
If yes, give a rationale for the use of deceptive or covert research (max 500 words)	

PREVIOUS EXPERIENCE
1 What previous experience of research involving human participants, relevant to this project, do you have? (max 500 words)
As a final year BSc Software Engineering student, I have experience in software development, requirements gathering, software testing, and user-centred system design through academic coursework and project work. I have received training on research ethics, academic integrity, confidentiality, informed consent, and responsible data



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handling. I have also conducted surveys and collected user feedback as part of previous academic assignments and software engineering activities.

2. Student projects only

What previous experience of research involving human participants, relevant to this project, does your supervisor have? (max 500 words)

Details of the supervisor's experience will be provided following confirmation from the supervisor.

POTENTIAL RISKS

Provide details of any risks you foresee related to the project and, for each risk, explain the methods you will use to reduce them. Ensure that you include details of measures that will be taken to ensure the safety of participants in the event that any identified risks arise.

Include details of risks to the participants, the researcher and the project as a whole.

NB: You must provide detail related to all risks identified in the previous question. Failure to do so could result in your application being returned to you.

(max 1000 words)

Data Privacy Risk (Participants): There is a minimal risk regarding the privacy of data collected during the survey and UAT. *Mitigation:* All survey and testing data will be fully anonymized. No personally identifiable information (PII) will be published in the final dissertation. Consent forms will be provided prior to any participation.

- **Data Security Risk (System):** The application itself handles user data and location information. *Mitigation:* The system uses secure token-based authentication (Laravel Sanctum), encrypts API communications, and utilizes fuzzed geographical coordinates to ensure user locations are masked during the matching process.
- **Project Completion Risk (Researcher):** AI model integration and PostgreSQL spatial extensions are complex. *Mitigation:* The system architecture includes "Context-Aware Graceful Degradation," meaning if the AI services fail or take too long, the system safely falls back to a manual filter-based search, ensuring the application remains functional and the project is completed on time.

When submitting your application, you **MUST** attach a copy of all information sheets, the consent/assent form(s) and the withdrawal of consent form.

An exemplar information sheet, exemplar participant consent form and exemplar participant withdrawal form are available via the research section of the Cardiff Met website (see section on Ethics Governance). These are based on good practice and will be useful in the majority of cases. However, it is recognised that in some cases a project will be subject to requirements from an external body. Use of these exemplars is therefore not obligatory.



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DECLARATION

I confirm that this project conforms with the [Cardiff Met Research Integrity & Governance Framework](#)

I confirm that I will abide by the Cardiff Met requirements regarding confidentiality and anonymity when conducting this project.

STUDENTS: I confirm that I will not disclose any information about this project without the prior approval of my supervisor.

Signature of the applicant:

Dilaksha Dissanayake

Date:

2026/06/10

FOR STUDENT PROJECTS ONLY

Name of supervisor:

Ms. Upeka Wijesinghe

Date: 28/06/2026

Signature of supervisor:

Research Ethics Committee use only

Decision reached:

[Click here to enter text.](#)

Project reference number: [Click here to enter text.](#)

Name: [Click here to enter text.](#)

Date: [Click here to enter a date.](#)

Details of any conditions upon which approval is dependant:

[Click here to enter text.](#)